Employee Payroll System

Java Swing Source Code

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import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.io.*;
import java.util.ArrayList;

// Employee class
class Employee
{
    int id;
    String name;
    double salary;
    String department;

    // Default constructor
    Employee()
    {
        id = 0;
        name = "";
        salary = 0;
        department = "";
    }

    // Parameterized constructor
    Employee(int id, String name, double salary, String department)
    {
        this.id = id;
        this.name = name;
        this.salary = salary;
        this.department = department;
    }

    // Copy constructor
    Employee(Employee e)
    {
        id = e.id;
        name = e.name;
        salary = e.salary;
        department = e.department;
    }

    // Calculate net salary
    double netSalary()
    {
        return salary + 3000 - 1000;
    }

    // Convert employee details into text
    String getData()
    {
        return id + "," + name + "," + salary + "," + department;
    }
}

// Main class
public class EmployeePayroll extends JFrame implements ActionListener
{
    JTextField idField, nameField, salaryField;
    JComboBox<String> deptBox;
    JTextArea area;

    JButton addBtn, saveBtn, loadBtn, viewBtn;

    ArrayList<Employee> list = new ArrayList<Employee>();

    // GUI constructor
    EmployeePayroll()
    {
        setTitle("Employee Payroll System");
        setSize(600, 450);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout(10, 10));

        JPanel form = new JPanel(new GridLayout(4, 2, 5, 5));

        form.add(new JLabel("Employee ID:"));
        idField = new JTextField();
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        form.add(idField);

        form.add(new JLabel("Employee Name:"));
        nameField = new JTextField();
        form.add(nameField);

        form.add(new JLabel("Basic Salary:"));
        salaryField = new JTextField();
        form.add(salaryField);

        form.add(new JLabel("Department:"));
        String[] departments = {"IT", "HR", "Sales", "Finance"};
        deptBox = new JComboBox<String>(departments);
        form.add(deptBox);

        add(form, BorderLayout.NORTH);

        JPanel buttons = new JPanel();

        addBtn = new JButton("Add Employee");
        saveBtn = new JButton("Save");
        loadBtn = new JButton("Load");
        viewBtn = new JButton("View All");

        buttons.add(addBtn);
        buttons.add(saveBtn);
        buttons.add(loadBtn);
        buttons.add(viewBtn);

        add(buttons, BorderLayout.CENTER);

        area = new JTextArea(10, 40);
        area.setEditable(false);

        add(new JScrollPane(area), BorderLayout.SOUTH);

        addBtn.addActionListener(this);
        saveBtn.addActionListener(this);
        loadBtn.addActionListener(this);
        viewBtn.addActionListener(this);

        setLocationRelativeTo(null);
        setVisible(true);
    }

    // Button actions
    public void actionPerformed(ActionEvent e)
    {
        try
        {
            // Add Employee
            if (e.getSource() == addBtn)
            {
                int id = Integer.parseInt(idField.getText().trim());
                String name = nameField.getText().trim();
                double salary = Double.parseDouble(salaryField.getText().trim());
                String dept = (String) deptBox.getSelectedItem();

                if (id <= 0 || name.isEmpty() ||
                    salary < 0 || Double.isNaN(salary) ||
                    Double.isInfinite(salary) ||
                    name.contains(","))
                {
                    JOptionPane.showMessageDialog(this,
                        "Enter valid employee details. Do not use comma in name.");
                    return;
                }

                // Check duplicate ID
                for (Employee emp : list)
                {
                    if (emp.id == id)
                    {
                        JOptionPane.showMessageDialog(this,
                            "This Employee ID already exists.");
                        return;
                    }
                }

                Employee emp = new Employee(id, name, salary, dept);

                // Use copy constructor
                Employee copy = new Employee(emp);
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    list.add(copy);

    JOptionPane.showMessageDialog(this,
        "Employee Added Successfully");

    area.setText("Net Salary: Rs. " + copy.netSalary());

    idField.setText("");
    nameField.setText("");
    salaryField.setText("");
}

// Save records
else if (e.getSource() == saveBtn)
{
    try (BufferedWriter bw = new BufferedWriter(
        new FileWriter("employees.txt")))
    {
        for (Employee emp : list)
        {
            bw.write(emp.getData());
            bw.newLine();
        }
    }

    JOptionPane.showMessageDialog(this,
        "Records Saved Successfully");
}

// Load records
else if (e.getSource() == loadBtn)
{
    ArrayList<Employee> temp = new ArrayList<Employee>();

    try (BufferedReader br = new BufferedReader(
        new FileReader("employees.txt")))
    {
        String line;

        while ((line = br.readLine()) != null)
        {
            String[] data = line.split(",", -1);

            if (data.length != 4)
            {
                throw new IOException("Invalid record in file");
            }

            Employee emp = new Employee(
                Integer.parseInt(data[0]),
                data[1],
                Double.parseDouble(data[2]),
                data[3]);

            temp.add(emp);
        }
    }

    list.clear();
    list.addAll(temp);

    JOptionPane.showMessageDialog(this,
        "Records Loaded Successfully");
}

// View all employees
else if (e.getSource() == viewBtn)
{
    area.setText("");

    if (list.isEmpty())
    {
        area.setText("No employee records found.");
    }

    for (Employee emp : list)
    {
        area.append(
            "Employee ID: " + emp.id +
            "\nName: " + emp.name +
            "\nDepartment: " + emp.department +
            "\nBasic Salary: Rs. " + emp.salary +
            "\nNet Salary: Rs. " + emp.netSalary() +
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                    "\n----------------------------\n");
                }
            }
        }
        catch (NumberFormatException ex)
        {
            JOptionPane.showMessageDialog(this,
                "Enter ID as a whole number and salary as a valid number.");
        }
        catch (IOException ex)
        {
            JOptionPane.showMessageDialog(this,
                "File not found or file has an error.");
        }
    }

    // Main method
    public static void main(String[] args)
    {
        new EmployeePayroll();
    }
}
```